

Task 68: Happy Number

Problem Description

Write an algorithm to determine if a number n is happy. A happy number is a number defined by the following process: Starting with any positive integer, replace the number by the sum of the squares of its digits. Repeat the process until the number equals 1 (where it will stay), or it loops endlessly in a cycle which does not include 1. Those numbers for which this process ends in 1 are happy. Return true if n is a happy number, and false if not.

Java Solution

```
class Solution {
    public boolean isHappy(int n) {
        Set<Integer> seen = new HashSet<>();
        while (n != 1 && !seen.contains(n)) {
            seen.add(n);
            n = getNext(n);
        }
        return n == 1;
    }

    private int getNext(int n) {
        int totalSum = 0;
        while (n > 0) {
            int d = n % 10;
            n = n / 10;
            totalSum += d * d;
        }
        return totalSum;
    }
}
```